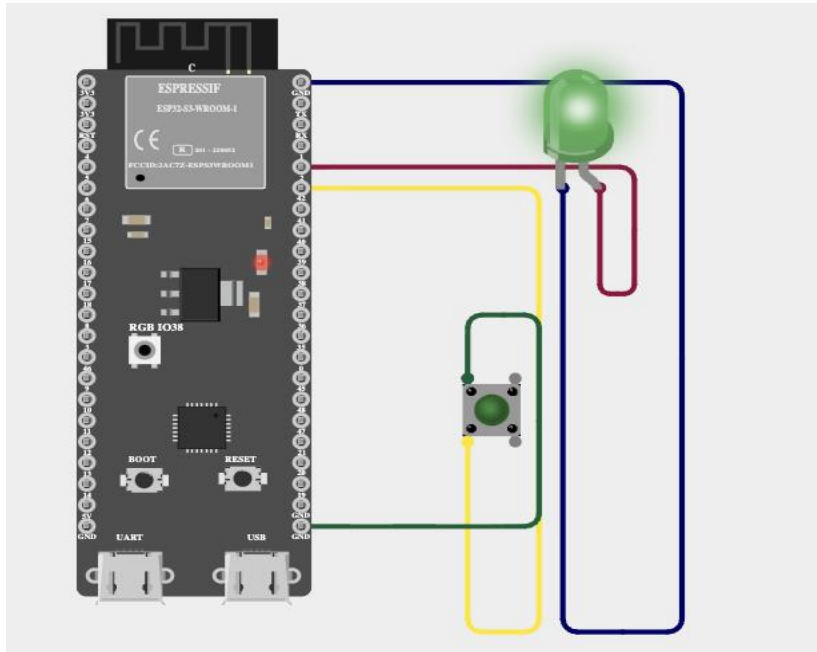


Embedded Computing Principles

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EXPERIMENT

LED Blinking & GPIO Control



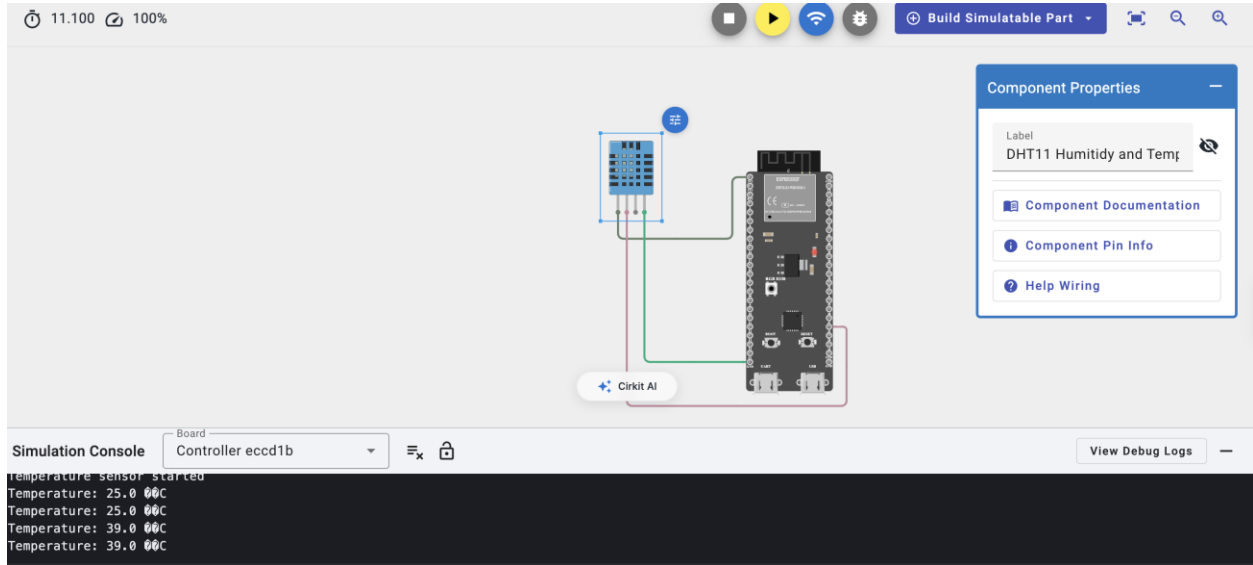
CODE:

```
const int LED_PIN = 2;

void setup() {
  pinMode(LED_PIN, OUTPUT);
}

void loop() {
  digitalWrite(LED_PIN, HIGH);
  delay(500);
  digitalWrite(LED_PIN, LOW);
  delay(500);
}
```

Temperature Monitoring



CODE:

```
#include <DHT.h>

const uint8_t TEMPERATURE_PIN = 21;

const uint8_t SENSOR_TYPE = DHT11;

DHT temperatureSensor(TEMPERATURE_PIN, SENSOR_TYPE);

void setup() {

  Serial.begin(115200);

  delay(500);

  temperatureSensor.begin();

  Serial.println("Temperature sensor started");

}

void loop() {

  // DHT11 needs a short interval between readings.

  delay(2000);

  float temperatureC = temperatureSensor.readTemperature();

  if (isnan(temperatureC)) {

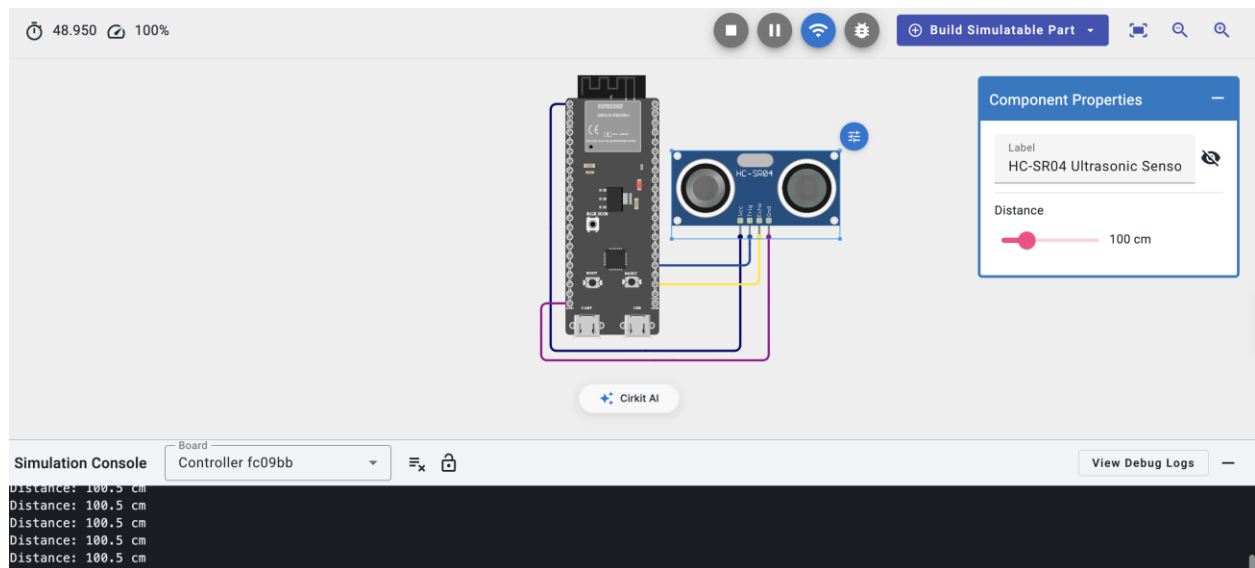
    Serial.println("Error: temperature reading failed");

  }

}
```

```
return;  
}  
Serial.print("Temperature: ");  
Serial.print(temperatureC, 1);  
Serial.println(" °C");  
}
```

Ultrasonic Distance Measurement



CODE:

```
const int TRIG_PIN = 21;

const int ECHO_PIN = 19;

void setup() {

  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);

  pinMode(ECHO_PIN, INPUT);

  digitalWrite(TRIG_PIN, LOW);

  delay(500);

  Serial.println("HC-SR04 distance monitor started");

}

void loop() {

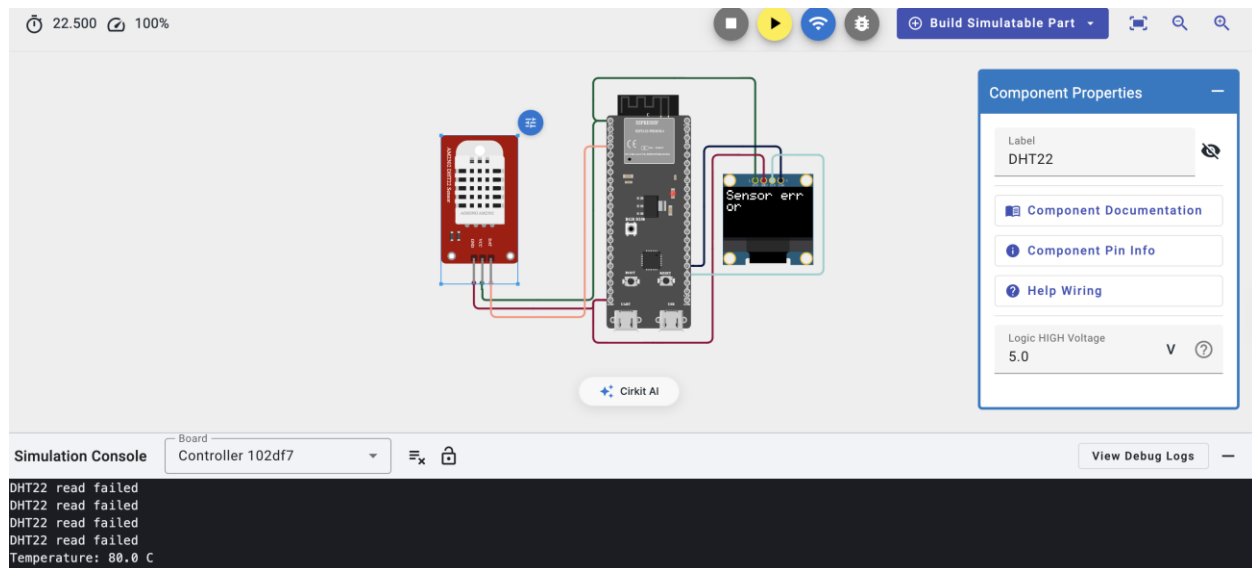
  // Send a 10 microsecond trigger pulse.

  digitalWrite(TRIG_PIN, LOW);

  delayMicroseconds(2);
```

```
digitalWrite(TRIG_PIN, HIGH);  
delayMicroseconds(10);  
digitalWrite(TRIG_PIN, LOW);  
// Measure the echo time; timeout prevents the loop from hanging.  
unsigned long duration = pulseIn(ECHO_PIN, HIGH, 30000);  
if (duration == 0) {  
  Serial.println("Distance: out of range or no echo");  
} else {  
  float distanceCm = duration / 58.0;  
  Serial.print("Distance: ");  
  Serial.print(distanceCm, 1);  
  Serial.println(" cm");  
}  
delay(500);  
}
```

OLED Display with I²C



```
#include <Wire.h>

#include <DHT.h>

#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define DHT_PIN 4

#define DHT_TYPE DHT22

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

#define OLED_RESET -1

#define OLED_ADDRESS 0x3C

DHT dht(DHT_PIN, DHT_TYPE);

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

void setup() {
  Serial.begin(115200);
  dht.begin();
  Wire.begin(21, 20); // SDA, SCL
```

```
if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDRESS)) {  
  Serial.println("OLED initialization failed");  
  while (true) {  
    delay(1000);  
  }  
}  
  
display.clearDisplay();  
display.setTextColor(SSD1306_WHITE);  
display.setTextSize(1);  
display.setCursor(0, 0);  
display.println("Temperature monitor");  
display.display();  
delay(1000);  
}  
  
void loop() {  
  float temperatureC = dht.readTemperature();  
  if (isnan(temperatureC)) {  
    Serial.println("DHT22 read failed");  
    display.clearDisplay();  
    display.setCursor(0, 0);  
    display.println("Sensor error");  
    display.display();  
  } else {  
    Serial.print("Temperature: ");  
    Serial.print(temperatureC, 1);  
    Serial.println(" C");  
    display.clearDisplay();
```

```
display.setTextSize(1);  
display.setCursor(0, 0);  
display.println("Temperature");  
display.setTextSize(2);  
display.setCursor(0, 22);  
display.print(temperatureC, 1);  
display.print(" C");  
display.display();  
}  
delay(2000); // DHT22 requires at least 2 seconds between readings  
}
```